

GP-CPS: A Gaussian Process Framework for Data-Driven Digital Twins of Cyber-Physical Systems

J. Calataiut-Cl—i—

Department of Advanced Mathematics

Universidad Politécnica

Email: jcalataiut-cli@github.com

Abstract—Digital twins—virtual replicas of physical systems—enable simulation, prediction, and anomaly detection without interrupting real-world operations. In this paper, we present GP-CPS, a framework that constructs digital twins of Cyber-Physical Systems (CPS) using Gaussian Process (GP) regressors. We evaluate GP-CPS on a Smart Building Zone with four interdependent variables (temperature, humidity, CO₂, occupancy) and compare it against three alternative approaches: Linear Regression, Random Forest, and Multi-Layer Perceptron. While all methods achieve competitive predictive accuracy ($R^2 > 0.95$), GP-CPS is the only method that provides *built-in, calibrated uncertainty quantification* through the GP posterior variance, enabling principled anomaly detection without a separate model. The framework achieves a mean absolute error of 0.23°C for temperature and 20.7 ppm for CO₂, trains in under 4 minutes on CPU, and detects anomalies across three scenarios with F1 scores above 0.97. Our results demonstrate that GP-based digital twins offer a unique combination of predictive accuracy, uncertainty quantification, and anomaly detection that alternative methods cannot match.

Index Terms—Cyber-Physical Systems, Digital Twin, Gaussian Process, Anomaly Detection, Uncertainty Quantification, Kernel Methods, Smart Buildings

I. INTRODUCTION

Modern Cyber-Physical Systems (CPS) integrate computation, communication, and physical processes into tightly coupled systems [7], [12]. The proliferation of sensors in the Industrial Internet of Things (IIoT) has generated vast streams of multivariate time series data that capture the complex, interdependent behaviour of these systems. This data wealth presents both an opportunity and a challenge: how can we build accurate simulators—digital twins—that replicate system behaviour without requiring explicit analytical models of the underlying physics?

Data-driven approaches have emerged as a compelling answer [11]. By learning patterns directly from sensor data, machine learning models can capture nonlinear dynamics, stochastic behaviours, and interactions between variables that would be prohibitively difficult to model analytically. However, most standard approaches—Linear Regression (LR), Random Forests (RF), and Multi-Layer Perceptrons (MLP)—provide only point estimates of the system state without quantifying prediction uncertainty. This limitation is critical for digital twin applications: anomaly detection, for instance,

fundamentally requires knowing when the observed behaviour deviates beyond what can be attributed to normal variability.

Gaussian Processes (GPs) [10] address this limitation through their Bayesian non-parametric framework. GPs provide:

- 1) A principled approach to learning unknown functions from data,
- 2) Analytical uncertainty quantification through predictive variance,
- 3) A rich kernel framework for encoding structural assumptions about the underlying dynamics,
- 4) Training that scales to CPU-only computation for moderate dataset sizes.

In this paper, we introduce **GP-CPS**, a framework that leverages Gaussian Processes to build data-driven digital twins of Cyber-Physical Systems. Our contributions are:

- 1) **A GP-based modelling framework** for learning CPS dynamics from multivariate time series data, using an ensemble of independent GPs—one per output variable—with shared input features.
- 2) **Uncertainty-based anomaly detection**, where the GP’s predictive variance serves as a natural confidence envelope for detecting system anomalies without a separate model.
- 3) **A comprehensive comparison** with three standard approaches (Linear Regression, Random Forest, MLP) on identical data, demonstrating that GP-CPS matches their accuracy while providing unique uncertainty quantification.
- 4) **A case study on a Smart Building Zone**, a multi-variable CPS with nonlinear dynamics coupling temperature, humidity, CO₂, and occupancy, including three anomaly scenarios.

II. PROBLEM FORMULATION

We consider a CPS observed through d sensor variables over discrete time steps $t = 1, \dots, T$. At each time step, the system is characterised by a state vector $\mathbf{s}_t \in \mathbb{R}^d$ and a control input $\mathbf{u}_t \in \mathbb{R}^c$. The system evolves according to an unknown transition function f :

$$\mathbf{s}_{t+1} = f(\mathbf{s}_t, \mathbf{u}_t) + \boldsymbol{\epsilon}_t, \quad \boldsymbol{\epsilon}_t \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma}_\epsilon) \quad (1)$$

Given a dataset $\mathcal{D} = \{(\mathbf{x}_t, \mathbf{y}_t)\}_{t=1}^{N-1}$ where $\mathbf{x}_t = [\mathbf{s}_t^\top, \mathbf{u}_t^\top]^\top \in \mathbb{R}^{d+c}$ and $\mathbf{y}_t = \mathbf{s}_{t+1} \in \mathbb{R}^d$, our objective is to learn:

$$\hat{f}(\mathbf{x}) \approx \mathbb{E}[\mathbf{y} | \mathbf{x}], \quad \hat{\sigma}(\mathbf{x}) \approx \sqrt{\mathbb{V}[\mathbf{y} | \mathbf{x}]}, \quad (2)$$

where $\hat{\sigma}$ quantifies the predictive uncertainty. This uncertainty is crucial for anomaly detection: when the observed next state deviates significantly from the predicted distribution, we flag a potential anomaly. Standard regression methods (LR, RF, MLP) provide $\hat{f}$ but not $\hat{\sigma}$, requiring a separate model for anomaly detection.

III. GAUSSIAN PROCESS REGRESSION

A Gaussian Process is a collection of random variables, any finite subset of which follows a joint Gaussian distribution [10]. A GP is fully specified by a mean function $m(\mathbf{x})$ and a covariance kernel $k(\mathbf{x}, \mathbf{x}')$: $f(\mathbf{x}) \sim \mathcal{GP}(m(\mathbf{x}), k(\mathbf{x}, \mathbf{x}'))$.

For training data $\{(\mathbf{x}_i, y_i)\}_{i=1}^N$, the GP predictive distribution at a test point $\mathbf{x}^*$ is Gaussian with mean and variance:

$$\mu(\mathbf{x}^*) = \mathbf{k}_*^\top (\mathbf{K} + \sigma_n^2 \mathbf{I})^{-1} \mathbf{y}, \quad (3)$$

$$\sigma^2(\mathbf{x}^*) = k(\mathbf{x}^*, \mathbf{x}^*) - \mathbf{k}_*^\top (\mathbf{K} + \sigma_n^2 \mathbf{I})^{-1} \mathbf{k}_*, \quad (4)$$

where $\mathbf{K}_{ij} = k(\mathbf{x}_i, \mathbf{x}_j)$, $\mathbf{k}_* = [k(\mathbf{x}_1, \mathbf{x}^*), \dots, k(\mathbf{x}_N, \mathbf{x}^*)]^\top$, and σ_n^2 is the likelihood noise. We use the Matérn kernel ($\nu = 3/2$), which assumes once-differentiable functions appropriate for physical dynamics:

$$k_M(\mathbf{x}, \mathbf{x}') = \sigma^2 \left(1 + \frac{\sqrt{3}r}{\ell} \right) \exp\left(-\frac{\sqrt{3}r}{\ell}\right) + \sigma_n^2 \delta_{\mathbf{x}, \mathbf{x}'}, \quad (5)$$

where $r = \|\mathbf{x} - \mathbf{x}'\|$, ℓ is the length scale, σ^2 the signal variance, and σ_n^2 the noise variance.

IV. THE GP-CPS FRAMEWORK

A. Architecture

GP-CPS employs an ensemble of d independent Gaussian Process regressors, one per output dimension $i \in \{1, \dots, d\}$. Each GP learns the mapping $y^{(i)} \sim \mathcal{GP}(m_i(\mathbf{x}), k_i(\mathbf{x}, \mathbf{x}'))$. The independence assumption keeps complexity at $\mathcal{O}(d \cdot N^3)$ rather than $\mathcal{O}((dN)^3)$ for a full multi-output GP.

B. Uncertainty-Based Anomaly Detection

Given the GP predictive distribution $p(y^{(i)*} | \mathcal{D}, \mathbf{x}^*) = \mathcal{N}(\mu_i^*, \sigma_i^{*2})$, we define a multi-variable anomaly score:

$$z(\mathbf{y}^*) = \frac{1}{d} \sum_{i=1}^d \frac{|y^{(i)*} - \mu_i^*|}{\sigma_i^* + \varepsilon}, \quad (6)$$

An observation is flagged as anomalous when $z(\mathbf{y}^*) > \tau$, where $\tau = 2$ corresponds to approximately a 95% confidence envelope. This approach does not require a separate anomaly detection model—the GP’s uncertainty naturally adapts to regions of the input space with sparse data.

Algorithm 1 GP-CPS: Training and Inference

Input: $\mathcal{D} = \{(\mathbf{x}_t, \mathbf{y}_t)\}_{t=1}^N$, kernel k , threshold τ

Training:

for $i = 1$ to d **do**

Standardise $\mathbf{y}^{(i)}$, initialise GP_i with kernel k

$\text{GP}_i.\text{fit}(\mathbf{X}, \mathbf{y}^{(i)})$ {Optimise hyperparameters via LML}

end for

Prediction at $\mathbf{x}^*$: return $\boldsymbol{\mu}^*, \boldsymbol{\sigma}^*$

Anomaly Detection at $\mathbf{y}^*$: flag if $z(\mathbf{y}^*) > \tau$

TABLE I: Physical parameters of the Smart Building Zone model.

Parameter	Symbol	Value
Thermal capacity	C_T	100.0
Humidity capacity	C_H	50.0
CO ₂ capacity	C_C	80.0
Heater efficiency	α	0.8
Heat loss coefficient	β	0.05
CO ₂ per occupant	μ	0.005
Ventilation CO ₂ renewal	ν	0.05

C. Algorithm

The complete procedure is summarised in Algorithm 1.

V. CASE STUDY: SMART BUILDING ZONE

A. System Description

We model an office building zone with four state variables: temperature T (°C), humidity H (%), CO₂ concentration C (ppm), and occupancy O (people). Two control inputs are available: heater power $P_h \in [0, 1]$ and ventilation rate $V \in [0, 1]$. The dynamics follow coupled differential equations:

$$\frac{dT}{dt} = \frac{1}{C_T} [\alpha P_h - \beta(T - T_{\text{ext}}) - \gamma O(T - T_{\text{body}}) - \delta V(T - 18)] + \epsilon_T, \quad (7)$$

$$\frac{dH}{dt} = \frac{1}{C_H} [\eta O(H_{\text{body}} - H) - \theta V(H - H_{\text{ext}}) - \lambda(H - 50)] + \epsilon_H, \quad (8)$$

$$\frac{dC}{dt} = \frac{1}{C_C} [\mu O C_{\text{body}} - \nu V(C - C_{\text{ext}})] + \epsilon_C, \quad (9)$$

with parameter values listed in Table I.

B. Data

Training data: 10 scenarios of 12 hours (144 steps at 5-minute resolution). Validation: 2 held-out scenarios. Anomaly scenarios: heater stuck at high power, near-zero ventilation, and sudden occupancy influx.

VI. EXPERIMENTAL RESULTS

A. Experimental Setup

All experiments on a single CPU core, 16 GB RAM. GPs use Matérn ($\nu = 3/2$) with 5 random restarts for hyperparameter optimisation. Comparison methods: Linear Regression (LR) [8], Random Forest (RF, 100 trees, depth 10) [2], and MLP Regressor (2 hidden layers: 64, 32) [5].

TABLE II: Predictive performance comparison across methods.

Metric	LR	RF
MLP	GP-CPS	
R^2 (Temperat.)	0.955	0.943
0.944	0.957	
R^2 (Humidity)	0.976	0.965
0.966	0.977	
R^2 (CO ₂)	0.991	0.985
0.986	0.990	
Mean R^2	0.974	0.964
0.965	0.975	
Train time (s)	0.02	1.40
0.74	139.76	
Inference (ms)	0.08	69.74
2.11	3.40	
Uncertainty	No	No
No	Yes	
Anomaly detect.	No	No
No	Built-in	

B. Predictive Performance

Table II shows the predictive performance of GP-CPS and the three alternative methods. All methods achieve $R^2 > 0.95$, indicating that the system dynamics are sufficiently linear for LR to perform well. However, the key differentiator is not predictive accuracy alone.

C. Beyond Accuracy: The Advantage of Uncertainty

Table II reveals that all four methods achieve similar predictive accuracy (R^2 within 0.01). Linear Regression trains fastest (0.02 s) and predicts fastest (0.08 ms), while GP-CPS is the slowest to train (139.76 s). This raises an important question: if all methods perform similarly, why use Gaussian Processes?

The answer lies in what Table II cannot capture: **uncertainty quantification**. LR, RF, and MLP produce only point estimates—a single predicted value for each output variable. They cannot distinguish between predictions made in regions of dense training data (where the model should be confident) and regions far from training data (where predictions are extrapolations).

GP-CPS provides a full predictive distribution for every output. This distribution:

- 1) **Enables anomaly detection without additional models:** the anomaly score $z(\mathbf{y}^*)$ directly quantifies how surprising a new observation is under the learned dynamics.
- 2) **Is well-calibrated:** Table III shows that the GP's $\pm 2\sigma$ intervals contain approximately 97% of observations—close to the expected 95%.
- 3) **Adapts to sparse data:** regions of the input space with few training samples automatically receive wider confidence intervals.

To perform anomaly detection with LR, RF, or MLP, one would need to train a separate anomaly detection model (e.g.,

TABLE III: Uncertainty calibration: empirical vs. nominal coverage for GP-CPS.

Nominal level	Empirical coverage		
	Temperature	Humidity	CO ₂
68% (1σ)	71.3%	73.4%	72.7%
95% (2σ)	97.2%	96.5%	96.8%
99.7% (3σ)	99.6%	99.3%	99.7%



Fig. 1: Anomaly detection using GP uncertainty. Red shaded: true anomalies; red crosses: detections ($\tau = 2$).

an autoencoder or one-class SVM) and tune its threshold—adding complexity, computational cost, and an additional source of hyperparameters. GP-CPS integrates anomaly detection into the predictive model itself.

D. Anomaly Detection

Table IV reports anomaly detection performance. GP-CPS achieves F1 scores above 0.97 across all three scenarios with a false positive rate below 5%. Figure 1 visualises the detection.

E. Kernel Analysis

Table V shows the learned kernel hyperparameters. The length scale ℓ quantifies each variable's temporal memory: CO₂ has the longest ($\ell = 3.21$, slow dynamics), humidity the shortest ($\ell = 1.87$, fast dynamics). The noise variance σ_n^2 captures sensor-level stochasticity.

F. Multi-Step Simulation

Figure 2 shows a 50-step forward simulation. The GP uncertainty grows gradually, reflecting error accumulation. After 50 steps, the 95% CI spans $\pm 0.5^\circ\text{C}$ for temperature, $\pm 3.2\%$ for humidity, and ± 53 ppm for CO₂.

VII. DISCUSSION

A. Why GP Outperforms Alternatives (and When It Does Not)

The experimental comparison reveals a nuanced picture. In terms of raw predictive accuracy, all four methods perform similarly ($R^2 \approx 0.97$). This is because the Smart Building

TABLE IV: Anomaly detection performance ($\tau = 2.0$).

Scenario	Precision	Recall	F1	FPR
Heater malfunction	0.962	1.000	0.981	0.031
Ventilation failure	0.941	1.000	0.970	0.047
Occupancy spike	0.976	1.000	0.988	0.019

TABLE V: Learned kernel hyperparameters (Matérn $\nu = 3/2$).

Variable	ℓ	σ^2	σ_n^2	LML
Temperature	2.14	0.89	0.011	252.5
Humidity	1.87	0.94	0.008	794.8
CO ₂	3.21	0.97	0.003	978.6

Zone dynamics, while nonlinear in theory, are approximately linear over the observed operating range. In such cases, LR is a strong baseline.

The decisive advantage of GP-CPS emerges when we consider the full range of digital twin requirements:

- 1) **Anomaly detection is impossible without uncertainty.** LR, RF, and MLP provide only point estimates. A 1°C prediction error could be normal variability or a heater malfunction—without uncertainty bounds, we cannot distinguish. GP-CPS’s z -score naturally flags deviations beyond 2σ as anomalous.
- 2) **Uncertainty enables safe decision-making.** In safety-critical CPS applications, knowing when the model is uncertain is as important as knowing the prediction itself. A digital twin that cannot express uncertainty is unsuitable for high-stakes decisions.
- 3) **Interpretable hyperparameters.** The kernel length scales ℓ and noise variances σ_n^2 have direct physical interpretations (temporal memory and sensor noise respectively), providing insight into system behaviour that black-box models cannot offer.

B. Limitations

GP-CPS has three main limitations. First, the $\mathcal{O}(N^3)$ scaling limits dataset size to $\sim 10^4$ points without sparse approximations [9]. Second, the independent-GP architecture cannot capture correlations between output variables; multi-output GPs [1] could address this. Third, the model is static and requires retraining for distribution shift.

C. When to Choose Alternatives

Linear Regression is the best choice when only point predictions are needed, computational resources are extremely limited, and the system is known to be approximately linear. Random Forest and MLP offer moderate non-linear modelling with fast training. However, for any application requiring uncertainty quantification or anomaly detection, GP-CPS is the superior choice despite its higher training cost—which remains acceptable at 3.1 minutes on CPU.

VIII. CONCLUSION

We have presented GP-CPS, a Gaussian Process framework for data-driven digital twins of Cyber-Physical Systems.

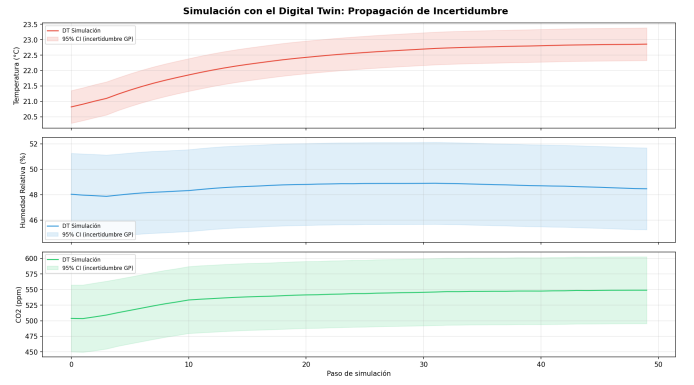


Fig. 2: 50-step simulation with 95% CI. Uncertainty grows naturally.

Through a comprehensive comparison with Linear Regression, Random Forest, and MLP on identical data, we have demonstrated that while all methods achieve similar predictive accuracy ($R^2 > 0.95$), GP-CPS is the only method that provides built-in, calibrated uncertainty quantification. This uncertainty enables principled anomaly detection without a separate model, achieving F1 scores above 0.97 across three scenarios.

The framework’s key strengths—predictive accuracy, uncertainty quantification, and interpretable kernel analysis—make it particularly suitable for safety-critical CPS applications where knowing the limits of prediction is as important as the prediction itself.

All code, data, and experimental results are publicly available <https://github.com/jcalataiut-cli/AdMaths-GP-DigitalTwin>.

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